

CLOUD SOC LAB

Azure NSG Security Rule Modification

INCIDENT RESPONSE REPORT

Incident Status: Investigated
Severity: Medium
Detection Platform: Microsoft Sentinel
Log Source: Azure Activity Logs
Data Table: AzureActivity
Test Environment: Azure Cloud SOC Lab
Date: 30th August 2026

1. Executive Summary

A potentially unauthorized modification to an Azure Network Security Group (NSG) security rule was detected using Microsoft Sentinel.

Azure Activity Logs were streamed to a Log Analytics Workspace and monitored using a custom Microsoft Sentinel Analytics Rule. The detection identified a successful NSG security rule modification performed by a separate TEST USER account.

The investigation identified:

- The user responsible for the modification is the same user who bruteforced an Azure Account
- The Azure operation performed
- The affected Network Security Group resource
- The specific security rule involved
- The success status of the operation

The incident demonstrated an end-to-end cloud SOC workflow including **log collection, detection engineering, incident creation, investigation, entity mapping, and automated incident handling.**

2. Lab Environment

Component	Technology
Cloud Platform	Microsoft Azure
SIEM/SOAR	Microsoft Sentinel
Log Collection	Azure Activity Logs

Component	Technology
Log Analytics Workspace	LAW-CLOUDSOC
Detection	Scheduled Analytics Rule
Automation	Sentinel Automation Rule
Workflow	Azure Logic Apps Playbook
Resource Group	RG-CloudSOC-Lab

3. Incident Scenario

A separate TEST USER account was used to simulate a potentially unauthorized modification to an Azure Network Security Group security rule.

The activity generated an Azure administrative event recorded in Azure Activity Logs.

The detected operation was:

Microsoft.Network/networkSecurityGroups/securityRules/write

The affected NSG security rule identified during investigation was:

Allow-RDP-Internet

This type of activity is security-sensitive because modifying NSG rules can expose cloud resources to unauthorized network access.

4. Detection Logic

A custom Microsoft Sentinel Scheduled Analytics Rule was created to monitor Azure Activity Logs for NSG security rule modifications.

Detection Query

AzureActivity

| where TimeGenerated > ago(1h)

| where OperationNameValue has "networkSecurityGroups/securityRules"

| project

TimeGenerated,

OperationNameValue,

Resource,

ResourceId,

Properties,

Caller

| order by TimeGenerated desc

The query monitored Azure administrative operations related to NSG security rules.

5. Key Evidence

The investigation identified the following important information.

Responsible User

The user responsible for the activity was identified through the:

Caller field. -> test-soc@zephyr[REDACTED]

Operation Performed

Microsoft.Network/networkSecurityGroups/securityRules/write

This confirmed that an NSG security rule was created or modified.

Activity Status

Success

The successful status confirmed that the configuration change was applied.

Affected Resource

The affected Azure resource was identified through the Azure Activity Log event properties.

Modified Security Rule

By investigating the Properties field and Azure resource entity path, the specific NSG rule was identified as:

Allow-RDP-Internet

This allowed the investigation to move from a general detection of:

NSG security rule modification detected

to a more actionable finding:

A specific user successfully modified the Allow-RDP-Internet NSG security rule.

6. Investigation Process

The SOC investigation followed these steps:

Step 1 — Incident Review

The Microsoft Sentinel incident was opened and reviewed.

The incident provided:

- Severity
 - Incident status
 - Related analytics rule
 - Timeline
 - Account entities
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Step 2 — Identify the Responsible User

The account entity associated with the incident was reviewed.

The Caller field in Azure Activity Logs was used to identify the account responsible for the modification.

Step 3 — Identify the Azure Operation

The OperationNameValue field was analyzed.

The operation identified was:

Microsoft.Network/networkSecurityGroups/securityRules/write

Step 4 — Investigate Event Properties

The Azure Activity Log Properties field was expanded.

The entity field contained the Azure resource path associated with the modification.

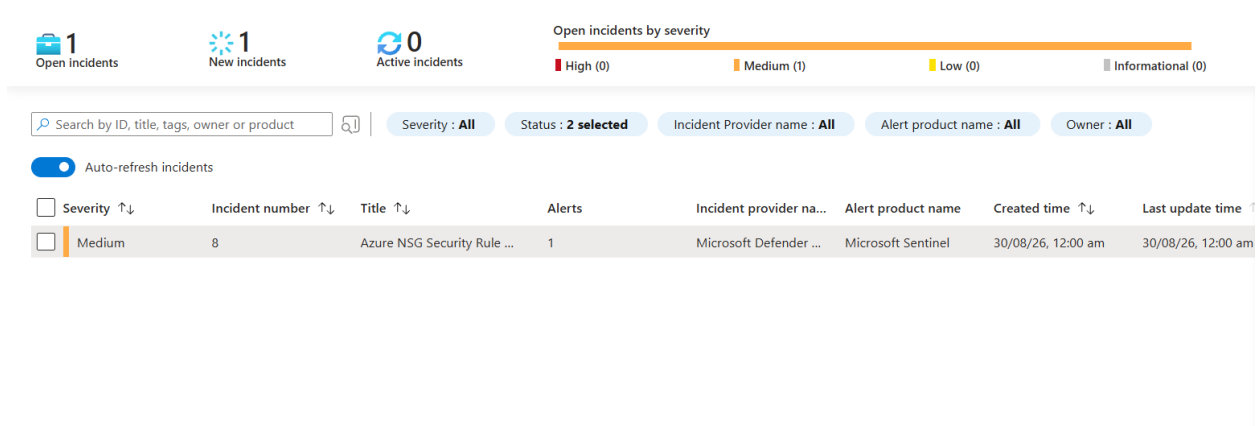
This provided visibility into the affected NSG and security rule.

Step 5 — Identify the Modified Rule

The resource entity path revealed the affected rule:

Allow-RDP-Internet

This was the key investigative finding.



7. Additional Investigation Query

The following query can be used to investigate NSG security rule modifications:

AzureActivity

| where TimeGenerated > ago(24h)

| where OperationNameValue has "networkSecurityGroups/securityRules"

| project

TimeGenerated,

Caller,

OperationNameValue,

Resource,

ResourceGroup,

ActivityStatusValue,

Properties

| order by TimeGenerated desc

To investigate activity performed by a specific user:

AzureActivity

| where TimeGenerated > ago(24h)

| where Caller contains "<USER>"

| project

TimeGenerated,

Caller,

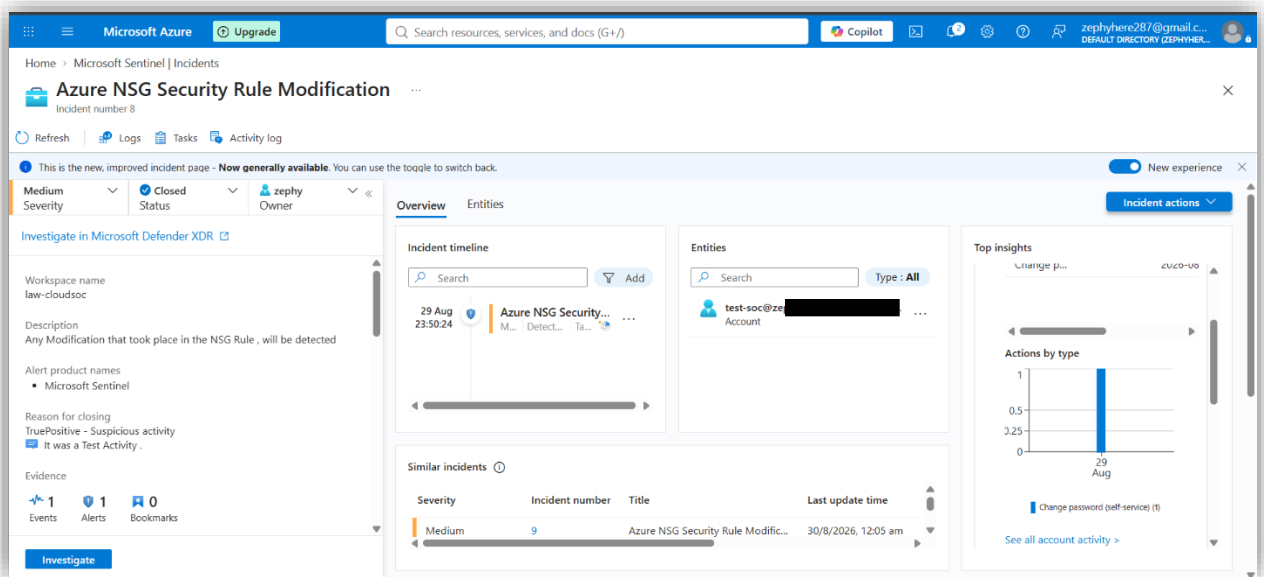
OperationNameValue,

Resource,

ResourceGroup,

ActivityStatusValue

| order by TimeGenerated desc



8. Incident Automation

A Microsoft Sentinel Automation Rule was configured for incidents generated by the NSG modification detection.

The automation workflow was designed to automatically process matching incidents and demonstrate Sentinel SOAR capabilities.

An Azure Logic Apps Playbook named:

NSG-Modification-Automated-Response

was also created as part of the automated response architecture.

The project demonstrated the integration between:

Analytics Rule



Sentinel Incident



Automation Rule



Automated Response Workflow

9. Security Impact

Unauthorized modification of NSG rules can result in significant security risks.

Potential impact includes:

- Exposure of management ports such as RDP
- Unauthorized internet access
- Increased risk of brute-force attacks
- Cloud infrastructure misconfiguration
- Bypass or weakening of network security controls
- Persistence through unauthorized firewall modifications

The **Allow-RDP-Internet rule** was considered security-sensitive because changes to RDP exposure could potentially increase the attack surface of Azure virtual machines.

10. MITRE ATT&CK Mapping

Technique	Description
T1562.007	Impair Defenses: Disable or Modify Cloud Firewall
T1078	Valid Accounts

Relevant Tactics

- Defense Evasion
 - Initial Access
 - Persistence
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11. Recommended Response Actions

In a production environment, the SOC team should:

1. Verify whether the NSG modification was authorized.
 2. Contact the responsible administrator or resource owner.
 3. Review the previous and current NSG rule configuration.
 4. Validate source IP ranges and exposed ports.
 5. Revert unauthorized security rule changes.
 6. Review additional Azure Activity Logs from the same user.
 7. Investigate related authentication activity.
 8. Restrict unnecessary public access to management ports.
 9. Apply least-privilege access controls for NSG modifications.
 10. Monitor for repeated configuration changes.
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12. Final Assessment

The simulated NSG security rule modification was successfully detected and investigated using Microsoft Sentinel.

The investigation successfully identified:

- **Who performed the modification** -> test-soc@zephyrhere287gmail.onmicrosoft.com
- **What Azure operation was performed** -> Modified NSG RULES
- **Whether the operation succeeded** -> Success
- **Which Azure resource was affected** -> NSG-CLOUD-SOC-LAB /Allow-RDP-Internet
- **Which NSG security rule was modified** -> Allow-RDP-Internet

The project successfully demonstrated a complete **Cloud SOC detection and investigation workflow**, from Azure log ingestion to incident generation and response automation.

Final Outcome

Detection: Successful

Incident Creation: Successful

User Identification: Successful

NSG Rule Identification: Successful

Investigation: Successful

Automation Configuration: Successful
